Mechanism Plausibility in Generative Agent-Based Modeling

Anonymous Author(s)

Abstract

Large language models (LLMs) can generate high-level phenomena without explicitly programmed rules. This capability has led to their adoption within agent-based models (ABMs) and social simulations. Recent research papers aim to test whether they are capable of generating different phenomena of interest, for example, human behavior on social media platforms or performance in game-theoretic scenarios.

However, capability, prediction, and explanation are different – drawing from the philosophy of science and mechanisms literature, *explanation* requires showing, to some degree, how a phenomenon is produced by related organized entities and activities. For modelers, describing the characteristics of an experiment or whether a simulation provides progress in capability (or explanation), can be particularly difficult without first being grounded in potentially distant research areas.

We integrate recent work on LLM-ABMs with contemporary philosophy of science literature and make two main contributions. First, we gather insights from modeling and mechanisms literature and use them to operationalize a definition of plausibility in a four-level scale. Our scale separates the evaluation of a model's generative sufficiency (ability to reproduce a phenomenon) from its mechanistic plausibility (how the phenomenon could be produced), and clarifies the distinct roles of different models, such as predictive and explanatory ones. We introduce this as the Mechanism Plausibility Scale. Second, we discuss the early wave of LLM-ABM research and find that papers often conflate evidence of Agent-level functionality with claims about emergent ABM-level phenomenon, relying on 'believability' metrics that focus on generative sufficiency. Our discussion section speaks on how these findings echo long-standing problems in classical ABM, historical harms caused by these issues, and broader ethical and epistemic concerns about using LLMs in modeling. Using the findings from our review, we offer the scale as a practical heuristic in the form of a checklist which can clarify how simulations at different levels of plausibility may be useful. We hope the activity of filling out the scale will help new modelers ground the epistemic contribution of their simulations.

CCS Concepts

• **Computing methodologies → Simulation evaluation**; **Model development and analysis**; **Modeling and simulation**; *Natural language processing*; • **Human-centered computing** → *Collaborative and social computing design and evaluation methods.*

Keywords

Agent-Based Modeling, Mechanisms, Generative Agents, Large Language Models, Philosophy of Science

This paper was prepared for PoliSim@CHI 2026: LLM Agent Simulation for Policy, a workshop at CHI 2026 (CHI Conference on Human Factors in Computing Systems), April 16, 2026, Barcelona, Spain.

1 Introduction

Developments in natural language processing have spurred interest in using large language models (LLMs) in social simulations [3, 26, 44], for example, extending the action space of agents in agent-based models (ABMs). It seems increasingly possible that product decisions, policymaking, and even research itself may be influenced by the outcome of such simulations [3, 35]. Today, a modeler [60] creating simulations with LLMs is capable of reproducing higher-level phenomena without explicitly programming the mechanisms that produce them. In a canonical agent-based model (ABM), the mechanisms underlying a phenomenon are operationalized and programmed by the modeler (e.g., for an economic agent, "if the price of resource A is X and my internal requirement B is at a threshold Y, then buy some amount of resource A"). These rules are typically based on some combination of assumptions, scientific theory, and empirical data. For instance, a programmer might read several social science papers to determine a particular distribution from which their agents will draw values that represent their preferences and attributes. Using LLMs in simulations offer the tantalizing promise that weights and biases obtained by training on social data may contain relevant distributional information about human behavior, allowing for richer representations of human subjects [3, 44, 56]. Critiques have also formed around models failing to capture the lived experiences of the human subjects they substitute [1].

When using LLMs in modeling social phenomena, we are left with a few puzzles: For a given simulation, did the results emerge from some correctly retrieved social knowledge encoded in the LLM's weights? Do our agents model the human behavior we are interested in? This could be the case, given that LLMs are trained on data describing real, human decision-making. Without improvements in the field of machine learning (ML) interpretability and data attribution, it could be the case that simulations incorporating LLMs are drawing on information that is irrelevant to the modeler's intent (sometimes referred to in the machine learning space as 'faithfulness' [47, 63]). In other words, we might produce a simulation that looks like it is *explaining* a social science phenomenon, but is just generating it through some other means, regardless of the 'how'. One can imagine that there are many ways in which a phenomenon can occur, and we are only interested in a particular one.

A recent review by Larooij et al. from April 2025 [32] surveyed and found that most studies of generative ABMs with LLMs do not acknowledge much of established work in the traditional simulation literature, or even have proper operational validity. One particular summary is that much of recent evaluation rests on some variant of believability, where human annotators are tasked with labeling whether or not they think the outputs of agent dialogues come from a human. On top of this, much work focuses whether or not a simulation or its LLM agents are even capable of reasonably producing a target behavior.

We connect work from the philosophy of science about what can be learned from idealized computer models such as ABMs. In this paper we explain how simulations can vary in their level of plausibility and introduce a set of criteria for categorizing simulations along our axis of interest. This is not to say that the value of a simulation is dictated purely by plausibility or the mechanistic understanding of a model; it is of general agreement that idealizations and abstractions are common, if not, necessary in building accepted models, or science may never move forward [7, 17, 46]. Simulations can vary in their level and type of claim, whether they are predictive, illustrative, exploratory, explanatory, etc. However, if a modeler wants to use their simulation to make any level of claim about how T might arise in actuality (explanatory), they must move beyond a purely phenomenal account. We present a checklist version of the scale in Section 4, motivated by dataset and model information checklists in past work [21, 42, 62].

2 Motivation and Related Work

2.1 Operationalization

In the philosophy of science, *phenomena* are defined to be stable patterns, regularities, or events that can be reliably inferred from data, and are the targets of explanation for scientific communities [9]. The patterns that qualify as 'phenomena' are scoped to the particular domain of inquiry [39], and may vary depending on the modeler's methodological choices or research question [41].

Originating from psychometric evaluation, hypothetical constructs are a theoretical attribute postulated to explain observed behavioral patterns, but are not directly observable itself [16, 37]. We often work with hypothetical constructs in order to characterize and reason about mental and social phenomena [50, 57]. To allow empirical measurements of these constructs, we create operational definitions: these are an explicit, unambiguous set of operations, protocols, or rules that are treated as equivalent to these abstract constructs within the bounds of the experiment, for the sake of falsifiable detection and experimental reproducibility [11]. The process of creating an operational definition for a particular concept is called the "operationalization" of the concept. Since our focus is on simulation models created by researchers, most researchers have particular phenomena that they would like the simulation to produce [24]. We refer to these phenomena of interest as T, the *target phenomena*, and we expand on this definition in Section 2.2.

2.2 Phenomenal Models and Generative Sufficiency

The term *phenomenal* comes from established usage in the scientific modeling field [30, 40], where it is used to describe models that attempt to produce the patterns describing a target phenomenon T, but do not necessarily contain information about the mechanism behind it. They may produce an accurate output without describing the relevant internal structure, therefore limiting their *explanatory* power.

One can imagine that it is not always practical to produce the target phenomenon. In the modeling field, the term *generative sufficiency* describes the level to which a model is able to accurately produce T [24]. Due to the black-box nature of deep neural networks and other practicality reasons, much of the emphasis in the traditional machine learning field is put on the generative sufficiency of different ML models–how accurately they are able to produce a target behavior. At the time of writing, the primary ways to evaluate LLMs revolve around measuring scores they achieve on some benchmark centered around human evaluation or fact-checking. This mentality may have spread to the LLM social simulation area, as initial projects in the space used similar evaluations to benchmark the realism of their simulation. For example, projects using 'believability' as a metric for their sufficiency in producing a target behavior [28, 34, 44, 45, 51, 58].

While generative sufficiency could be appropriate for exploratory or illustrative settings, the goals of a model may not be limited to just reproducing the target behavior; One may want to test unknown counterfactual scenarios using their simulation once they have built up a reasonable model. Problematically, these simulations are evaluated based off of their generative sufficiency and then used to test interventions as if there are plausible mechanisms [19, 27]. Phenomenal models cannot be used to test counterfactual scenarios because they lack the relevant internal causal structure. In order for a model to move past being purely phenomenal, it needs to reveal something about how T is produced: the mechanisms behind it.

2.3 Mechanisms

Mechanisms literature has seen a rise in discussion in the past two decades, primarily in the philosophy-of-science and neuroscience fields [13, 22, 38]. Glennan gives a minimal definition for mechanisms in his text [22]: *"A mechanism for a phenomenon consists of entities (or parts) whose activities and interactions are organized so as to be responsible for the phenomenon."* Pragmatically, in our discussion of agent-based models and computer simulation, this might include how the agents, environment, and update rules function to produce T. A mechanism is involved in the causal process of T, not just correlated, and hypothesizing about them is the first step towards explanation of the complex system of interest. This hypothesis may take the form of a mapping, which details what parts of the simulation correspond to mechanisms behind T. In our scale the addition of this mapping is what distinguishes a purely phenomenal model from a model that presents plausible candidate mechanisms for the ones behind T.

To explain the mechanisms behind a phenomenon is to explain how the phenomenon is produced (falsifiably). Once some level of description of the mechanisms behind T is produced, the model is beyond a purely phenomenal account. To be clear, an explanation does not need to constitute every detail down to the atomic level; it can use an adequate level of abstraction or idealization to fit the use case of the modeler [14, 55]. Lower-level mechanisms beyond the 'Agent-level' could be further explored, but also may be stubbed at the modeler's adequate level of abstraction.

We note that mechanisms cannot be identified in isolation, and therefore the target phenomena need to be operationalized before identifying any mechanisms. Craver suggests, "mechanistic explanations can fail because one has tried to explain a fictitious phenomenon, because one has mischaracterized the phenomenon, and because one has characterized the phenomenon to be explained only partially." [15] Mechanisms are not just 'static' concepts, they are functions that are defined relative to a phenomenon. Its identity, boundaries, and relevance are all defined by the specific outcome it is supposed to explain [13, 23]. Therefore, as we will see later, in our Mechanism Plausibility Scale the operationalization precedes the hypothesis.

2.4 Mechanisms vs Prediction

The focus on the productive process of a phenomenon distinguishes mechanisms from predictivism or correlation. One can use a barometer reading to predict weather, but the changing air pressure it measures is not the mechanism that produces it. In causal inference, this is the difference between observational and interventional questions [48, 49]. A predictive model is appropriate for questions such as "given these initial conditions, what outcomes are likely?" Without plausible mechanisms, however, a model's predictive outputs are usually only appropriate to the extent that they can be validated against observed outcomes. For scenarios that have been or can be empirically tested, this validation may suffice [53]. But for novel interventions, mechanisms provide the basis for reasoning about whether the model's outputs are acceptable.

This distinction between explanation and prediction is well established and known to be easily conflated [53]. The two goals require different criteria for model evaluation, different relationships to the underlying data-generating process, etc. An explanatory model aims to test causal hypotheses about the process producing T; a predictive model aims to produce accurate forecasts of new observations, and may do so using variables that have no/weak causal relationship to the outcome. Conflating the two is a category error that appears in both classical statistics and, as we argue, in early LLM-ABM work.

2.5 Plausibility

The definition of plausibility can be vague, subjective and is often treated as a qualitative property; From the Stanford Encyclopedia of Philosophy: *"To say that a hypothesis is plausible is to convey that it has epistemic support: we have some reason to believe it, even prior to testing."* [8].

In this paper, we operationalize plausibility in ABMs based off of its standing on our scale (presented in Section 3), which encapsulates factors such as how a simulation's components are operationalized, the type of evidence used to justify its parameters, the model's relationship to any hypotheses the modeler is presenting, etc. In particular, we are interested in if a model is a faithful representation of the modeler's intent. As mentioned previously, scholars recently publishing in the LLM-ABM space have used believability/plausibility metrics like human annotation to support their simulations. We find that the task of identifying what the results of these evaluations actually show may be elusive for even seasoned and capable researchers when it is applied to LLM simulation, giving us the primary motivation for developing our scale.

3 A Mechanism Plausibility Scale

Now that we have distinguished explanatory models from predictive and illustrative ones, we introduce an axis for models as plausible explanations.

Craver: *"For those interested in building plausible simulations, it will not suffice for simulation S simply to reproduce the input–output mapping of target phenomenon T. The model is further constrained by what is known about the internal machinery by which the inputs are transformed into outputs. It is possible, for example, to simulate human skills at multiplication with two sticks marked with logarithmic scales; but that is not how most humans multiply."* [14]

If we want a simulation to be a *plausible* representation for how T is created, it is not sufficient to just reproduce T – the simulation mechanisms must be adequately close to being a proxy for how T may actually be generated [29, 61].

We formalize a **model** as a four-tuple $M = (S, T, I, E)$: **Simulation** (S), **Target** phenomenon (T), modeler **Intent** (I), and **Evidence** (E) (these terms will be further defined below). We use this four-tuple to create a corresponding four-level **Mechanism Plausibility Scale**. Models climb our scale as more components of M become falsifiable and relevant in explaining its mechanisms, as well as being overall more faithful in representing the modeler's intentions.

3.1 Level 0

A Level 0 model is a 'toy' simulation or sandbox with no specified modeling goal. It consists of a Simulation S, which is set of procedures, code, and update rules that generate outputs but lacks a clearly defined phenomenon to explain. Since mechanisms are defined relative to a phenomenon (sometimes referred to as *Glennan's Law*) [13, 23], a model without a target cannot have mechanisms. We place models that lack explicit operationalization of a target T unintentionally in level 0 as well.

A classical example of a level 0 simulation is Conway's Game of Life [20]. While it generates complex emergent behavior from simple rules, it was conceived as a mathematical curiosity without a target phenomenon T to model or explain.

3.2 Level 1

To reach Level 1, a model must add an operationalized target T. T is the system or phenomenon the simulation is trying to represent or generate.

In mechanisms literature, models that do not convey the underlying mechanism are said to be *phenomenal*, their purpose being pattern reproduction rather than creating hypotheses with their simulation [13, 14]. Thus, simulations at level 1 are phenomenal models. They also have an operationally defined T and are considered generatively sufficient if its simulation S can produce the operationalized patterns of T. It makes no claims about explanation; S exists to reproduce T and does not explain the mechanisms behind it. To put it another way, they are 'hard-coded' simulations that produce a set of data points which match an operational pattern T.

Recently, Level 1 simulations with LLM agents have been used to explore the capabilities of different language models in cooperation, games, and other environments to benchmark how different LLMs perform in such abstract settings. For example, there exists a multitude of work on placing LLMs in game-theoretic scenarios to see how they act or perform [2, 12, 25, 31, 36]. In these scenarios, researchers may have created T retroactively after the simulation, an appropriate process in experimentation.

The research questions of these simulations often follow the lines of "can some LLM agents produce behaviors $\{x_0, x_1, \ldots, x_m\}$" and are questions of generative sufficiency, rather than related to *why* a behavior was produced. More generally, a lot of work uses multi-agent LLM simulations to probe the capabilities of LLMs themselves, such as their capacity for cooperation or their inherent biases. The goal of these simulations is to characterize the LLM agents' capabilities, not to necessarily explain a specific real-world social dynamic. In the survey of recent generative ABM literature by Larooij et al. [32], we see that many projects use believability as their primary evaluation metric in this way, which we argue is an assessment of generative sufficiency rather than explanatory power.

The existence of a level 1 bucket also helps to flag if an LLM-based simulation could be 'cheating'. It is entirely feasible to produce LLM agents that return the outputs which produce T through the manipulation of prompts. The behaviors agents can be heavily influenced by prompt engineering; an engineered prompt that produces a desired behavior is perfectly acceptable for a level 1 phenomenal model. However, if an explanatory claim is being made, it can become important to clarify in the model's intent I whether the prompt is an 'artifact' that forces the correct output, or an intentional abstraction of a real-world mechanism. Without this clarification, a model may be phenomenally ambiguous.

3.3 Level 2

Simulations with a plausibility of level 2 move beyond reproducing T to proposing a hypothesis for how it could possibly be generated. This is achieved when modeler specifies their Intent I, which includes a mapping (sometimes called a 'model key' [24]) that connects the components and activities in S to the proposed mechanisms responsible for generating T. By creating a mapping, one states a hypothesis about how the possible mechanisms of T are related to the simulation code.

Modeling literature often refers to these post-phenomenal simulations as 'how-possibly' simulations [10, 24, 64] or 'logical possibilities' [4]. One distinguishment of level 2 from level 1 simulations is that they provide a basis for reasoning about counterfactual scenarios, given a hypothesis about T's mechanisms encoded through I. For example, one could reason about how T might change if those mechanisms were different.

Notably, the modeler's intent I can differentiate two models with the same simulation structure S and target T.

3.4 Level 3

A simulation with plausibility level 3 attempts to ground its components in Evidence E. E is data used to support or constrain the model's construction, parameterization, or validation. This evidence can inform the design of the simulation S; the operationalization of the target T; or the justification for the mapping in I. E could also come in the form of further constrained experiments, for example, ablations or sensitivity analyses that reinforce the prescribed hypotheses contained in the mapping. A modeler might select initial parameters based off of some observed values from census data, survey results, or prior empirical studies (falsifiably). For example, initializing an agent's political beliefs based on real-world polling data from a specific region might constitute a piece of evidence E.

For example, Epstein et al.'s Artificial Anasazi model [18] could be considered a level 3 simulation. It simulated the societal dynamics of ancestral Puebloan peoples in Long House Valley, Arizona. The model is directly supported by anthropological data, as its annual agricultural productivity is recreated using paleoenvironmental data from tree rings, soil analysis, geology, etc. In addition, the agents' behavior such as nutritional needs, household size, and reproduction rates are based on real anthropological studies of Puebloan peoples and similar agricultural societies, modeled into simple rules.

3.5 Clarifications on the Role of Levels

The value of a simulation is not purely dictated by explanation or the mechanistic understanding of a model; It is of general agreement among scientists and philosophers that idealizations are useful, if not, necessary in building models [7, 46]. The goal of the scale is not to rank simulations by 'quality', but to clarify the kind of epistemic contribution each one can provide, as well as if they are an accurate representation of the modeler's intent. For example, level 0 simulations like cellular automata can demonstrate that emergent behavior can arise from simple rules, which sets the ground for new simulation paradigms. Level 2 simulations can be used to generate "how-possibly" hypotheses which are falsifiable at the mechanisms level, for example, Schelling's model of segregation.

4 Applying the Mechanism Plausibility Scale

4.1 The Mechanism Plausibility Scale Heuristic

We draw on existing frameworks for reporting on machine learning datasets and model deployments [21, 42, 62] and present a checklist for using the Mechanism Plausibility Scale. While hypothesis testing and operationalization are long-standing frameworks in science, the novelty of LLM-ABMs may have led researchers to struggle with putting out artifacts that still satisfy these foundations. In Appendix A, Figure 1 we present the heuristic and in Appendix B we follow 4 running examples, one for each level in the scale.

5 Discussion

5.1 Reflections on the State of LLM Social Simulation

The current state of LLM social simulation research is that many contributions focus primarily on demonstrating their simulations are capable of producing some target phenomena (generative sufficiency). The addition of LLMs in social simulation seemed to move us further up the "generative sufficiency scale", allowing agents to access a larger action space. This focus on generative sufficiency is reflected in the systematic review by Larooij et al., where 22 out of 35 surveyed LLM social simulation papers used 'believability' as their primary validation metric [32], judged by humans or LLMs.

So, until breakthroughs in validation happen via interpretability, new methodologies, etc., how can we make current simulations useful for policy or other applied settings? Li et al. tackle this question in a year-long human co-design of simulations with their university's emergency preparedness team [35]. In their survey, the surveyed policymakers seemed to show immediate skepticisms towards any models' predictive abilities, even if the agents exhibited believable behavior. Instead, the simulations seemed to help them more as a brainstorming tool. For example, when a simulation's dynamics were identified to be wrong, it would resurface the policymakers' tacit knowledge and allow them to explicitly list out important concerns that should be considered, for example, wheelchair ramps in evacuation settings. This has echoes in work done by Park et al. [45], where 'false' simulated social media platforms helped designers identify and prototype solutions to potential problems before they came up in a real deployed setting. The preparedness team began to trust the simulations more once the simulations started to align with real-world scenarios, when the authors tested it against their institution's real-world graduation commencement setting, and when the models' outputs began to align with their personal intuitions. Once the policymakers saw that the simulations sufficiently generated behavior (T) that matched outcomes based on their experience and evidence E (level 3), they were willing to entertain the 'how-possibly' outcomes generated by the simulation's higher-level, abstracted mechanisms.

5.2 Broader Ethical and Epistemic Concerns

The problems related to the conflation of generative sufficiency with mechanistic plausibility are methodological, and the difficulties in determining a simulation's usefulness is epistemic. However, there are broader ethical concerns about the use of LLMs in social simulation that also warrant consideration.

Regarding whether LLMs should serve as proxies for human subjects in the first place, Agnew et al. [1] examine proposals to substitute human research participants with LLM surrogates and find that such proposals conflict with values relating to representation, inclusion, and understanding of human subjects. Replacing participants with LLMs may disregard the relationship between researcher and subject existing in prior human subject research. When an LLM generates text that resembles survey responses or social behavior, it is not directly from the experience of a live, present individual. Furthermore, they identify the problem of "value lock-in", also referenced by Weidinger et al. [59]. LLMs encode the norms and attitudes present in their training data at a particular point in time. Related empirical work supports this; language models exhibit degraded performance in time periods not represented in their training corpus [33].

5.3 Historical Issues and Damages of Poor ABM Specification

While the Mechanism Plausibility Scale was motivated by recent challenges posed by LLM-ABM, the ideas are not specific to ABMs with LLMs; The literature surrounding well-motivated, sound ABM design in general is a long-standing discussion [5, 32, 43, 54–56]. Importantly, we demonstrate how understanding a model's limits is not only important to the modeler herself, but also to its end users.

Squazzoni et al. [54] note that during the COVID-19 pandemic, a team at the Imperial College of London reported that results from their model projected "a huge number of people would die in Britain unless severe policy measures were taken". The results of their model and interventions were quickly adopted and implemented by the UK government, and advised governments of countries like the US and France in their attempts to minimize the damages caused by the virus. However, because of underspecification on what the model was adequate for, the model erroneously affected the policies of many countries, namely, being used in counterfactual scenarios when further peer analysis of the model showed it may only have been adequate for illustrative purposes. Moreover, the simulation code was not made public, even later at the time of Squazzoni et al.'s publication.

Axelrod's iterated Prisoner's Dilemma (PD) simulations are another well-known problematic case. In an adapted script on "The Evolution of Cooperation" [6], Axelrod asserts that many real-world scenarios such as arms races, nuclear proliferation, and crisis bargaining are instances of the iterated Prisoner's Dilemma, and that advice to players of the game theoretic scenario might serve as advice to national leaders. In response, Northcott and Alexandrova [43] observe that despite the enormous attention devoted to the PD (over 16,000 articles since 1960), it has largely failed to explain phenomena of social scientific interest.

Arnold (sharply) observes a broader pattern in the modeling tradition [5]: over thirty years of Repeated PD simulations produced practically no successful empirical applications, yet this failure has been largely ignored. He accounts, firstly, the "justificatory narratives" modelers use after scrutiny, which is retreating to claims that the model is merely heuristic or exploratory without specifying the limits of that exploration. Secondly, modelers arguing that all models rely on simplification, a defense that, as Arnold notes, only holds when the causal factors a model isolates are empirically discernible from the other factors at work in the target system. When they are not, the simplification cannot be tested.

We felt it appropriate to reiterate these issues under our scale and point to related work, especially with the growing interest in simulation using LLMs.

6 Conclusion and Limitations

In this paper we connect contemporary mechanisms and philosophy of science literature with agent-based modeling and LLM social simulation. We present the Mechanism Plausibility Scale, a heuristic that classifies simulations into levels based on the falsifiability and existence of components S, T, I, E and offer a practical checklist. Through a review of recent LLM-ABM papers we confirm the existence of common category errors between Agent-level and ABM-level validation and under-specified models. We connect these problems with existing issues in ABM and highlight historical harms caused in high-stakes scenarios. The main focus of the paper, ultimately, remains grounding multiple disciplines in common language and bringing these issues to attention.

References

[1] William Agnew, A. Stevie Bergman, Jennifer Chien, Mark Díaz, Seliem El-Sayed, Jaylen Pittman, Shakir Mohamed, and Kevin R. McKee. 2024. The illusion of artificial inclusion. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–12. doi:10.1145/3613904.3642703 arXiv:2401.08572 [cs].

[2] Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. 2025. Playing repeated games with Large Language Models. *Nature Human Behaviour* (May 2025). doi:10.1038/s41562-025-02172-y arXiv:2305.16867 [cs].

[3] Jacy Reese Anthis, Ryan Liu, Sean M. Richardson, Austin C. Kozlowski, Bernard Koch, James Evans, Erik Brynjolfsson, and Michael Bernstein. 2025. LLM Social Simulations Are a Promising Research Method. doi:10.48550/arXiv.2504.02234 arXiv:2504.02234 [cs].

[4] Eckhart Arnold. 2013. Simulation Models of the Evolution of Cooperation as Proofs of Logical Possibilities. How Useful Are They? *Etica E Politica* 15, 2 (2013), 101–138. https://philarchive.org/rec/ARNSMO Publisher: University of Trieste, Department of Philosophy.

[5] Eckhart Arnold. 2015. How Models Fail: A Critical Look at the History of Computer Simulations of the Evolution of Cooperation. In *Collective Agency and Cooperation in Natural and Artificial Systems*, Catrin Misselhorn (Ed.). Springer International Publishing, Cham, 261–279. doi:10.1007/978-3-319-15515-9_14

[6] Robert Axelrod. [n. d.]. The Evolution of Cooperation*. ([n. d.]). https://ee.stanford.edu/~hellman/Breakthrough/book/pdfs/axelrod.pdf

[7] N. Emrah Aydinonat. 2024. The puzzle of model-based explanation. In *The Routledge Handbook of Philosophy of Scientific Modeling* (1 ed.). Routledge, London, 177–192. doi:10.4324/9781003205647-16

[8] Paul Bartha. 2024. Analogy and Analogical Reasoning. In *The Stanford Encyclopedia of Philosophy* (fall 2024 ed.), Edward N. Zalta and Uri Nodelman (Eds.). Metaphysics Research Lab, Stanford University. https://plato.stanford.edu/archives/fall2024/entries/reasoning-analogy/

[9] James Bogen and James Woodward. 1988. Saving the Phenomena. *The Philosophical Review* 97, 3 (1988), 303–352. jstor:2185445 doi:10.2307/2185445

[10] Alisa Bokulich. 2014. How the Tiger Bush Got its Stripes: 'How Possibly' vs. 'How Actually' Model Explanations. *The Monist* 97, 3 (July 2014), 321–338. doi:10.5840/monist201497321

[11] P. W. (Percy Williams) Bridgman. 1927. *The Logic of Modern Physics.* The Macmillan Company.

[12] Anthony Costarelli, Mat Allen, Roman Hauksson, Grace Sodunke, Suhas Hariharan, Carlson Cheng, Wenjie Li, and Arjun Yadav. 2024. GameBench: Evaluating Strategic Reasoning Abilities of LLM Agents. doi:10.48550/arXiv.2406.06613 arXiv:2406.06613 [cs] version: 1.

[13] Carl Craver, James Tabery, and Phyllis Illari. 2024. Mechanisms in Science. In *The Stanford Encyclopedia of Philosophy* (fall 2024 ed.), Edward N. Zalta and Uri Nodelman (Eds.). Metaphysics Research Lab, Stanford University. https://plato.stanford.edu/archives/fall2024/entries/science-mechanisms/

[14] Carl F. Craver. 2006. When mechanistic models explain. *Synthese* 153, 3 (Dec. 2006), 355–376. doi:10.1007/s11229-006-9097-x

[15] Carl F. Craver. 2009. *Explaining the Brain.* Oxford University Press.

[16] Lee J. Cronbach and Paul E. Meehl. 1955. Construct validity in psychological tests. *Psychological Bulletin* 52, 4 (1955), 281–302. doi:10.1037/h0040957 Place: US Publisher: American Psychological Association.

[17] Catherine Z. Elgin. 2004. True Enough. *Philosophical Issues* 14 (2004), 113–131. https://www.jstor.org/stable/3050623 Publisher: [Wiley, Ridgeview Publishing Company].

[18] Joshua M. Epstein. 2006. *Generative Social Science: Studies in Agent-Based Computational Modeling* (stu - student edition ed.). Princeton University Press. http://www.jstor.org/stable/j.ctt7rxj1

[19] Chen Gao, Xiaochong Lan, Zhihong Lu, Jinzhu Mao, Jinghua Piao, Huandong Wang, Depeng Jin, and Yong Li. 2025. *S^3: Social-network Simulation System with Large Language Model-Empowered Agents.* arXiv:2307.14984 [cs] doi:10.48550/arXiv.2307.14984

[20] Martin Gardner. 1970. Mathematical Games. *Scientific American* 223, 4 (1970), 120–123. https://www.jstor.org/stable/24927642 Publisher: Scientific American, a division of Nature America, Inc..

[21] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. 2021. Datasheets for Datasets. doi:10.48550/arXiv.1803.09010 arXiv:1803.09010 [cs].

[22] Stuart Glennan. 2017. *The New Mechanical Philosophy.* Oxford University Press, Oxford.

[23] Stuart S. Glennan. 1996. Mechanisms and the nature of causation. *Erkenntnis* 44, 1 (Jan. 1996), 49–71. doi:10.1007/BF00172853

[24] Claudius Graebner. 2018. How to Relate Models to Reality? An Epistemological Framework for the Validation and Verification of Computational Models. *Journal of Artificial Societies and Social Simulation* 21, 3 (2018), 8.

[25] Fulin Guo. 2023. GPT in Game Theory Experiments. doi:10.48550/arXiv.2305.05516 arXiv:2305.05516 [econ].

[26] John J. Horton. 2023. Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus? doi:10.48550/arXiv.2301.07543 arXiv:2301.07543 [econ].

[27] Wenyue Hua, Lizhou Fan, Lingyao Li, Kai Mei, Jianchao Ji, Yingqiang Ge, Libby Hemphill, and Yongfeng Zhang. 2024. War and Peace (WarAgent): Large Language Model-based Multi-Agent Simulation of World Wars. doi:10.48550/arXiv.2311.17227 arXiv:2311.17227 [cs].

[28] Zhao Kaiya, Michelangelo Naim, Jovana Kondic, Manuel Cortes, Jiaxin Ge, Shuying Luo, Guangyu Robert Yang, and Andrew Ahn. 2023. Lyfe Agents: Generative Agents for Low-Cost Real-Time Social Interactions. arXiv:2310.02172 [cs] doi:10.48550/arXiv.2310.02172

[29] David Michael Kaplan and Carl F. Craver. 2011. The Explanatory Force of Dynamical and Mathematical Models in Neuroscience: A Mechanistic Perspective*. *Philosophy of Science* 78, 4 (2011), 601–627. doi:10.1086/661755 Publisher: [The University of Chicago Press, Philosophy of Science Association].

[30] Kendrick N. Kay. 2018. Principles for models of neural information processing. *NeuroImage* 180 (Oct. 2018), 101–109. doi:10.1016/j.neuroimage.2017.08.016

[31] Benjamin Kempinski, Ian Gemp, Kate Larson, Marc Lanctot, Yoram Bachrach, and Tal Kachman. 2025. Game of Thoughts: Iterative Reasoning in Game-Theoretic Domains with Large Language Models. In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS '25).* International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 1088–1097.

[32] Maik Larooij and Petter Törnberg. 2025. Do Large Language Models Solve the Problems of Agent-Based Modeling? A Critical Review of Generative Social Simulations. doi:10.48550/arXiv.2504.03274 arXiv:2504.03274 [cs].

[33] Angeliki Lazaridou, Adhiguna Kuncoro, Elena Gribovskaya, Devang Agrawal, Adam Liska, Tayfun Terzi, Mai Gimenez, Cyprien de Masson d'Autume, Tomas Kocisky, Sebastian Ruder, Dani Yogatama, Kris Cao, Susannah Young, and Phil Blunsom. 2021. Mind the Gap: Assessing Temporal Generalization in Neural Language Models. doi:10.48550/arXiv.2102.01951 arXiv:2102.01951 [cs].

[34] Huao Li, Yu Quan Chong, Simon Stepputtis, Joseph Campbell, Dana Hughes, Michael Lewis, and Katia Sycara. 2023. Theory of Mind for Multi-Agent Collaboration via Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing.* 180–192. doi:10.18653/v1/2023.emnlp-main.13 arXiv:2310.10701 [cs].

[35] Yuxuan Li, Sauvik Das, and Hirokazu Shirado. 2025. What Makes LLM Agent Simulations Useful for Policy? Insights From an Iterative Design Engagement in Emergency Preparedness. doi:10.48550/arXiv.2509.21868 arXiv:2509.21868 [cs].

[36] Yuxuan Li and Hirokazu Shirado. 2025. Spontaneous Giving and Calculated Greed in Language Models. doi:10.48550/arXiv.2502.17720 arXiv:2502.17720 [cs].

[37] Kenneth MacCorquodale and Paul E. Meehl. 1948. On a Distinction between Hypothetical Constructs and Intervening Variables. *Psychological Review* 55, 2 (1948), 95–107. doi:10.1037/h0056029

[38] Peter Machamer, Lindley Darden, and Carl F. Craver. 2000. Thinking about Mechanisms. *Philosophy of Science* 67, 1 (2000), 1–25. https://www.jstor.org/stable/188611 Publisher: [The University of Chicago Press, Philosophy of Science Association].

[39] Michela Massimi. 2022. Perspectival Ontology: Between Situated Knowledge and Multiculturalism. *The Monist* 105, 2 (March 2022), 214–228. doi:10.1093/monist/onab032

[40] Michael D. Mauk. 2000. The potential effectiveness of simulations versus phenomenological models. *Nature Neuroscience* 3, 7 (July 2000), 649–651. doi:10.1038/76606 Publisher: Nature Publishing Group.

[41] James W. McAllister. 1997. Phenomena and Patterns in Data Sets. *Erkenntnis (1975-)* 47, 2 (1997), 217–228. jstor:20012798 doi:10.1023/A:1005387021520

[42] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. 2019. Model Cards for Model Reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency.* 220–229. doi:10.1145/3287560.3287596 arXiv:1810.03993 [cs].

[43] Robert Northcott and Anna Alexandrova. 2015. Prisoner's Dilemma Doesn't Explain Much. In *The Prisoner?s Dilemma. Classic philosophical arguments.*, Martin Peterson (Ed.). Cambridge University Press, 64–84. https://philarchive.org/rec/NORPDD

[44] Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative Agents: Interactive Simulacra of Human Behavior. doi:10.48550/arXiv.2304.03442 arXiv:2304.03442 [cs].

[45] Joon Sung Park, Lindsay Popowski, Carrie Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2022. Social Simulacra: Creating Populated Prototypes for Social Computing Systems. *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Oct. 2022), 1–18. doi:10.1145/3526113.3545616 Conference Name: UIST '22: The 35th Annual ACM Symposium on User Interface Software and Technology ISBN: 9781450393201 Place: Bend OR USA Publisher: ACM.

[46] Wendy S. Parker. 2020. Model Evaluation: An Adequacy-for-Purpose View. *Philosophy of Science* 87, 3 (July 2020), 457–477. doi:10.1086/708691

[47] Debjit Paul, Robert West, Antoine Bosselut, and Boi Faltings. 2024. Making Reasoning Matter: Measuring and Improving Faithfulness of Chain-of-Thought Reasoning. arXiv. doi:10.48550/ARXIV.2402.13950 Version Number: 4.

[48] Judea Pearl. 2009. *Causality* (2 ed.). Cambridge University Press, Cambridge. doi:10.1017/CBO9780511803161

[49] Judea Pearl and Dana Mackenzie. 2018. *The book of why: The new science of cause and effect* (1 ed.). Basic Books, Inc., USA.

[50] Axel Pichler and Nils Reiter. 2022. From Concepts to Texts and Back: Operationalization as a Core Activity of Digital Humanities. *Journal of Cultural Analytics* 7, 4 (Dec. 2022). doi:10.22148/001c.57195

[51] Siyue Ren, Zhiyao Cui, Ruiqi Song, Zhen Wang, and Shuyue Hu. 2024. Emergence of Social Norms in Generative Agent Societies: Principles and Architecture.

doi:10.48550/arXiv.2403.08251 arXiv:2403.08251 [cs].

[52] Thomas C. Schelling. 1969. Models of Segregation. *The American Economic Review* 59, 2 (1969), 488–493. https://www.jstor.org/stable/1823701 Publisher: American Economic Association.

[53] Galit Shmueli. 2010. To Explain or to Predict? *Statist. Sci.* 25, 3 (Aug. 2010). doi:10.1214/10-STS330

[54] Flaminio Squazzoni, J. Gareth Polhill, Bruce Edmonds, Petra Ahrweiler, Patrycja Antosz, Geeske Scholz, Emile Chappin, Melania Borit, Harko Verhagen, Francesca Giardini, and Nigel Gilbert. 2020. Computational Models That Matter During a Global Pandemic Outbreak: A Call to Action. *JASSS - The Journal of Artificial Societies and Social Simulation* 23, 2 (March 2020). doi:10.18564/jasss.4298

[55] Samarth Swarup. 2019. Adequacy: What Makes a Simulation Good Enough?. In *2019 Spring Simulation Conference (SpringSim)*. 1–12. doi:10.23919/SpringSim.2019.8732895

[56] Loïs Vanhée, Melania Borit, Peer-Olaf Siebers, Roger Cremades, Christopher Frantz, Önder Gürcan, František Kalvas, Denisa Reshef Kera, Vivek Nallur, Kavin Narasimhan, and Martin Neumann. 2025. Large Language Models for Agent-Based Modelling: Current and possible uses across the modelling cycle. doi:10.48550/arXiv.2507.05723 arXiv:2507.05723 [cs] version: 1.

[57] Elina Vessonen. 2021. Conceptual engineering and operationalism in psychology. *Synthese* 199, 3 (Dec. 2021), 10615–10637. doi:10.1007/s11229-021-03261-x

[58] Lei Wang, Jingsen Zhang, Hao Yang, Zhi-Yuan Chen, Jiakai Tang, Zeyu Zhang, Xu Chen, Yankai Lin, Hao Sun, Ruihua Song, Xin Zhao, Jun Xu, Zhicheng Dou, Jun Wang, and Ji-Rong Wen. 2025. User Behavior Simulation with Large Language Model-based Agents. *ACM Trans. Inf. Syst.* 43, 2 (Jan. 2025), 55:1–55:37. doi:10.1145/3708985

[59] Laura Weidinger, Jonathan Uesato, Maribeth Rauh, Conor Griffin, Po-Sen Huang, John Mellor, Amelia Glaese, Myra Cheng, Borja Balle, Atoosa Kasirzadeh, Courtney Biles, Sasha Brown, Zac Kenton, Will Hawkins, Tom Stepleton, Abeba Birhane, Lisa Anne Hendricks, Laura Rimell, William Isaac, Julia Haas, Sean Legassick, Geoffrey Irving, and Iason Gabriel. 2022. Taxonomy of Risks posed by Language Models. In *2022 ACM Conference on Fairness Accountability and Transparency*. ACM, Seoul Republic of Korea, 214–229. doi:10.1145/3531146.3533088

[60] Michael Weisberg. 2007. Who Is a Modeler? *The British Journal for the Philosophy of Science* 58, 2 (2007), 207–233. https://www.jstor.org/stable/30115224 Publisher: [Oxford University Press, The British Society for the Philosophy of Science].

[61] Michael Weisberg. 2013. *Simulation and Similarity: Using Models to Understand the World*. Oxford University Press.

[62] Michael Winikoff, John Thangarajah, and Sebastian Rodriguez. 2025. A Scoresheet for Explainable AI. doi:10.48550/arXiv.2502.09861 arXiv:2502.09861 [cs].

[63] Gal Yona, Roee Aharoni, and Mor Geva. 2024. Can Large Language Models Faithfully Express Their Intrinsic Uncertainty in Words? (2024). doi:10.48550/ARXIV.2405.16908 Publisher: arXiv Version Number: 2.

[64] Dunja Šešelja. 2023. Agent-Based Modeling in the Philosophy of Science. In *The Stanford Encyclopedia of Philosophy* (winter 2023 ed.), Edward N. Zalta and Uri Nodelman (Eds.). Metaphysics Research Lab, Stanford University. https://plato.stanford.edu/archives/win2023/entries/agent-modeling-philscience/

Received 20 February 2026; accepted 18 March 2026

A The Mechanism Plausibility Scale

Mechanism Plausibility Scale Heuristic (S, T, I, E)

This scale grades the model's potential as a plausible explanation for the target phenomenon. While not every model's goal is to be explanatory, it is important to clarify when it is appropriate.

Level 0: Simulation (S) **–** ***Sandbox/Toy Model***

- ☐ **Simulation.** Is the simulation (S) defined, including environments, agents, and update rules?

The requirements for Level 0 are relatively minimal; It essentially shares the same requirements for something to be considered an ABM or simulation, and mainly exists to distinguish it from the other levels. Level 0 simulations are often toys or demonstrations of new simulation paradigms (e.g., Conway's Game of Life, a demo for a new simulation framework/technique).

Level 1: Target Phenomenon (T) **–** ***Phenomenal Model***

- ☐ **Defined Target Phenomenon.** Is a target phenomenon (T) operationalized (e.g. as statistical patterns, human annotations/observations, etc.)?
- ☐ **Generative Sufficiency.** Can the simulation (S) successfully generate the patterns described in (T)?
- ☐ **Reproducibility.** Does the reproducibility of the simulation (seeds, API versions, consideration of proprietary prompt injections or version changes, inherent stochasticity, etc.) match the reproducibility goals of the modeler or the field they are working in (sensitivity analysis requirements, etc.)?

The Level 1 requirement checks if there are explicit conditions that determine if the phenomena have happened and if the model produces the specified phenomena. Level 1 simulations often show that something (T) is possible or achievable using the entities and activities of the simulation (e.g. Can LLM agents solve games, pass theory of mind, benchmarks, etc.).

Level 2: Intent & Mapping (I) **–** ***How-Possibly Model***

- ☐ **Simulation Contribution.** Is the simulation's use case understood (e.g., predictive, exploratory, illustrative, explanatory, etc.)?
- ☐ **Falsifiable Hypotheses.** Does there exist a hypothesis for how the target phenomenon T arises from components in the model?
- ☐ **Mechanism Mapping.** Is there an Intent I (implicit or explicit mapping) which connects components of the simulation (S) to the hypothesized 'real-world' mechanisms of (T)?

Level 2 is a check for if the modelers have explicitly proposed how the phenomena (or parts of the phenomena) of interest are produced using the components of their simulation. This is important to make the model falsifiable.

Level 3: Evidence (E) **–** ***Plausible Model (if validated)***

- ☐ **Evidence Exists.** Is the model supported by some evidence E?
- ☐ **Relevance.** Is E directed towards the claims made in I? Could E, in principle, disconfirm the hypotheses in I?

Adding any E to get a simulation to Level 3 is straightforward; The difficulty lies in the model to remain valid, as the E becomes additional falsifiable parts of the model. What is considered acceptable E is bounded by the standards of the domain the model is in.

Conclusion: Based on the checklist, the model is classified as a Level [X] model.

Figure 1: The Mechanism Plausibility Scale in checklist form.

B Examples

B.1 Level 0 Example

Conway's Game of Life

This scale grades the model's potential as a plausible explanation for the target phenomenon. While not every model's goal is to be explanatory, it is important to clarify when it is appropriate.

Level 0: Simulation (S) – *Sandbox/Toy Model*

☑ **Simulation.** Is the simulation (S) defined, including environments, agents, and update rules?

The rules of the game are detailed: At each timestep, *"1. Every counter with two or three neighboring counters survives for the next generation. 2. Each counter with four or more neighbors dies (is removed) from overpopulation. Every counter with one neighbor or none dies from isolation. Each empty cell adjacent to exactly three neighbors–no more, no fewer–is a birth cell. A counter is placed on it at the next move."*

Level 1: Target Phenomenon (T) – *Phenomenal Model*

☐ **Defined Target Phenomenon.** Is a target phenomenon (T) operationalized (e.g. as statistical patterns, human annotations/observations, etc.)?
☐ **Generative Sufficiency.** Can the simulation (S) successfully generate the patterns described in (T)?
☑ **Reproducibility.** Does the reproducibility of the simulation (seeds, API versions, consideration of proprietary prompt injections or version changes, inherent stochasticity, etc.) match the reproducibility goals of the modeler or the field they are working in (sensitivity analysis requirements, etc.)?

While Conway selected his rules to try and make the behavior of populations in his sim unpredictable, for the reader there is no operationalized pattern that the simulation is intended to reproduce, his simulation described as a "solitaire" [20].

Level 2: Intent & Mapping (I) – *How-Possibly Model*

☐ **Simulation Contribution.** Is the simulation's use case understood (e.g., predictive, exploratory, illustrative, explanatory, etc.)?
☐ **Falsifiable Hypotheses.** Does there exist a hypothesis for how the target phenomenon T arises from components in the model?
☐ **Mechanism Mapping.** Is there an Intent I (implicit or explicit mapping) which connects components of the simulation (S) to the hypothesized 'real-world' mechanisms of (T)?

Due to the lack of T, the simulation does not describe any mechanisms. Mechanisms are defined relative to a phenomenon (see 'Glennan's Law' [13, 23]).

Level 3: Evidence (E) – *Plausible Model (if validated)*

☐ **Evidence Exists.** Is the model supported by some evidence E?
☐ **Relevance.** Is E directed towards the claims made in I? Could E, in principle, disconfirm the hypotheses in I?

Since E requires T and I, the model does not pass the Level 3 requirement.

Conclusion: Based on the checklist, the model is classified as a Level 0 Toy model.

Figure 2: Example for Level 0: The Mechanism Plausibility Scale applied to an implementation of Conway's Game of Life [20].

B.2 Level 1 Example

Game Theory LLM Benchmarking Sim

This scale grades the model's potential as a plausible explanation for the target phenomenon. While not every model's goal is to be explanatory, it is important to clarify when it is appropriate.

Level 0: Simulation (S) – *Sandbox/Toy Model*

☑ **Simulation.** Is the simulation (S) defined, including environments, agents, and update rules?

S includes the codebase for the simulation, which is an implementation of games such as Prisoner's Dilemma, Texas Hold'em, Staghunt, etc. played by different LLM Agents against each other.

Level 1: Target Phenomenon (T) – *Phenomenal Model*

☑ **Defined Target Phenomenon.** Is a target phenomenon (T) operationalized (e.g. as statistical patterns, human annotations/observations, etc.)?
☑ **Generative Sufficiency.** Can the simulation (S) successfully generate the patterns described in (T)?
☑ **Reproducibility.** Does the reproducibility of the simulation (e.g. seeds, API versions, consideration of proprietary prompt injections or version changes, inherent stochasticity) match the reproducibility goals of the modeler or the field they are working in (sensitivity analysis requirements, etc.)?

We aim to benchmark the behavior of different LLMs to see if they will solve game-theoretic scenarios, and if they produce optimal or well-known game-theoretic behavior (e.g., Tit-for-Tat, Nash Equilibrium); their execution of these strategies we will consider T. After running the simulation, we find that models with a larger parameter space tend to exhibit more aggressive behavior.

Level 2: Intent & Mapping (I) – *How-Possibly Model*

☐ **Simulation Contribution.** Is the simulation's use case understood (e.g., predictive, exploratory, illustrative, explanatory, etc.)?
☐ **Falsifiable Hypotheses.** Does there exist a hypothesis for how the target phenomenon T arises from components in the model?
☐ **Mechanism Mapping.** Is there an Intent I (implicit or explicit mapping) which connects components of the simulation (S) to the hypothesized 'real-world' mechanisms of (T)?

We do not suggest the mechanisms behind why some models play different styles of games than others. However, our findings do find a correlation between the aggressiveness of players and their parameter count.

Level 3: Evidence (E) – *Plausible Model (if validated)*

☐ **Evidence Exists.** Is the model supported by some evidence E?
☐ **Relevance.** Is E directed towards the claims made in I? Could E, in principle, disconfirm the hypotheses in I?

Since E requires I, the model does not pass the Level 3 requirement.

Conclusion: Based on the checklist, the model is classified as a Level 1 Phenomenal model.

Figure 3: Example for Level 1: The Mechanism Plausibility Scale applied to a fabricated game theory paper.

B.3 Level 2 Example

Example Schelling's Model of Segregation

This scale grades the model's potential as a plausible explanation for the target phenomenon. While not every model's goal is to be explanatory, it is important to clarify when it is appropriate.

Level 0: Simulation (S) – *Sandbox/Toy Model*

☑ **Simulation.** Is the simulation (S) defined, including environments, agents, and update rules?

S is defined as a grid of agents who move to adjacent empty spots if the percentage of their own color neighbors falls below a certain threshold (formalized in full paper).

Level 1: Target Phenomenon (T) – *Phenomenal Model*

☑ **Defined Target Phenomenon.** Is a target phenomenon (T) operationalized (e.g. as statistical patterns, human annotations/observations, etc.)?
☑ **Generative Sufficiency.** Can the simulation (S) successfully generate the patterns described in (T)?
☑ **Reproducibility.** Is the simulation reproducible?

The target T is the emergence of macro-level clustering, which represents residential segregation. The simulation is generatively sufficient as it always produces segregated neighborhoods.

Level 2: Intent & Mapping (I) – *How-Possibly Model*

☑ **Simulation Contribution.** Is the simulation's use case understood (e.g., predictive, exploratory, illustrative, explanatory, etc.)?
☑ **Falsifiable Hypotheses.** Does there exist a hypothesis for how the target phenomenon T arises from components in the model?
☑ **Mechanism Mapping.** Is there an Intent I (implicit or explicit mapping) which connects components of the simulation (S) to the hypothesized 'real-world' mechanisms of (T)?

Our Intent I is to demonstrate the feasibility of the hypothesis that extreme individual prejudice is not necessary for macro-segregation to occur. Mild preferences for similar neighbors are a sufficient mechanism. The agents' movement rules map to a simplification of the real-world mechanism of residents relocating based on neighborhood composition.

Level 3: Evidence (E) – *Plausible Model (if validated)*

☐ **Evidence Exists.** Is the model supported by some evidence E?
☐ **Relevance.** Is E directed towards the claims made in I? Could E, in principle, disconfirm the hypotheses in I?

The threshold parameters in our model are abstract and not derived from specific empirical data (e.g. census surveys). It proposes a "how-possibly" mechanism.

Conclusion: Based on the checklist, the model is classified as a Level 2 How-Possibly model.

Figure 4: Example for Level 2: The Mechanism Plausibility Scale applied to Schelling's Model of Segregation [52]

B.4 Level 3 Example

Example Anasazi Model (Epstein et al.)

This scale grades the model's potential as a plausible explanation for the target phenomenon. While not every model's goal is to be explanatory, it is important to clarify when it is appropriate.

Level 0: Simulation (S) – *Sandbox/Toy Model*

☑ **Simulation.** Is the simulation (S) defined, including environments, agents, and update rules?

S is an agent-based model that simulates the Long House Valley environment using rules for annual agricultural productivity, and defines agent behaviors based on specific rules for nutritional needs, household size, and reproduction rates. The details of these parameters, as well as the simulation code can be found in our GitHub repository.

Level 1: Target Phenomenon (T) – *Phenomenal Model*

☑ **Defined Target Phenomenon.** Is a target phenomenon (T) operationalized (e.g. as statistical patterns, human annotations/observations, etc.)?
☑ **Generative Sufficiency.** Can the simulation (S) successfully generate the patterns described in (T)?
☑ **Reproducibility.** Is the simulation reproducible?

The target T is the historical population dynamics and settlement patterns of the Long House Valley from 800 AD to 1350 AD. The simulation reproduces the population crash and abandonment of the valley.

Level 2: Intent & Mapping (I) – *How-Possibly Model*

☑ **Simulation Contribution.** Is the simulation's use case understood (e.g., predictive, exploratory, illustrative, explanatory, etc.)?
☑ **Falsifiable Hypotheses.** Does there exist a hypothesis for how T arises?
☑ **Mechanism Mapping.** Is there an Intent I (implicit or explicit mapping) which connects components of the simulation (S) to the hypothesized 'real-world' mechanisms of (T)?

The intent (I) maps the simulation's rules (agricultural yield vs. caloric needs) to the real-world mechanisms of the environmental factors affecting population growth. Therefore it hypothesizes that environmental shifts were the primary driver.

Level 3: Evidence (E) – *Plausible Model (if validated)*

☑ **Evidence Exists.** Is the model supported by some evidence E?
☑ **Relevance.** Is E directed towards the claims made in I? Could E, in principle, disconfirm the hypotheses in I?

The model is constrained by empirical Evidence E, which includes agricultural productivity is reconstructed using paleoenvironmental data from tree rings, soil analysis, and geology. Each agent's nutritional needs and household sizes are derived from anthropological studies of Puebloan peoples.

Conclusion: Based on the checklist, the model is classified as a Level 3 Plausible model.

Figure 5: Example for Level 3: The Mechanism Plausibility Scale applied to the Artificial Anasazi Model [18]